PSet 1: The Price of Housing

```
library(tidyverse)

data_file <- "fmr2027.csv"

# fips and state have leading zeros
fmr <- read_csv(
  data_file,
  col_types = cols(fips = col_character(), state = col_character())
)

fmr <- fmr |>
  mutate(
    metro = factor(metro, labels = c("nonmetro", "metro")),
    # fips ending in 99999 is a whole county, otherwise a town
    record_type = if_else(str_ends(fips, "99999"), "county", "sub-county")
  )

# 2 bedroom rows only
fmr_2 <- filter(fmr, bedrooms == 2)

# one row per HUD area and bedroom size
areas <- fmr |>
  group_by(hud_area_code, hud_area_name, metro, bedrooms) |>
  summarise(
    fmr = first(fmr),
    pop = sum(pop2024),
    n_records = n(),
    .groups = "drop"
  )
areas_2 <- filter(areas, bedrooms == 2)

# areas that cross state lines are kept once per state
state_areas <- fmr |>
  distinct(stusps, hud_area_code, metro, bedrooms, fmr)
```

```
state_counts <- fmr_2 |>
  group_by(stusps) |>
  summarise(records = n(), `HUD areas` = n_distinct(hud_area_code)) |>
  pivot_longer(-stusps, names_to = "unit", values_to = "count") |>
  mutate(stusps = fct_reorder(stusps, count, .fun = max))

ggplot(state_counts, aes(count, stusps)) +
  geom_line(aes(group = stusps), color = "grey70") +
  geom_point(aes(color = unit)) +
  scale_x_log10() +
  labs(x = "count", y = "stusps", caption = "Figure 1") +
  theme_minimal(12)
```

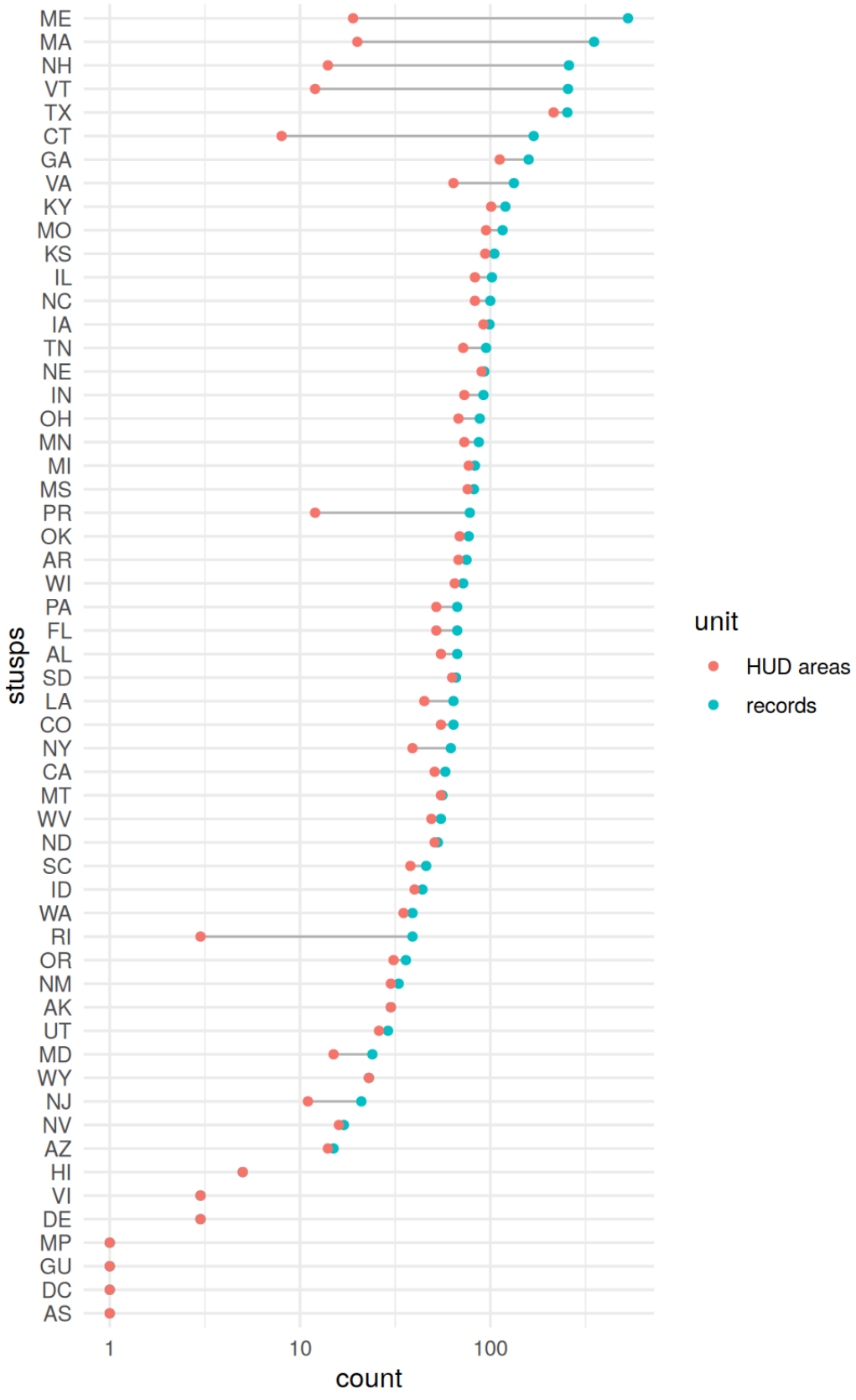


Figure 1

```
ggplot(areas, aes(fmr)) +
  geom_histogram(binwidth = 100, boundary = 0) +
  facet_grid(rows = vars(bedrooms), labeller = label_both) +
  scale_x_continuous(labels = scales::label_comma()) +
```

```
  labs(x = "fmr ($)", y = "count (HUD areas)", caption = "Figure 2") +
  theme_minimal(12)
```

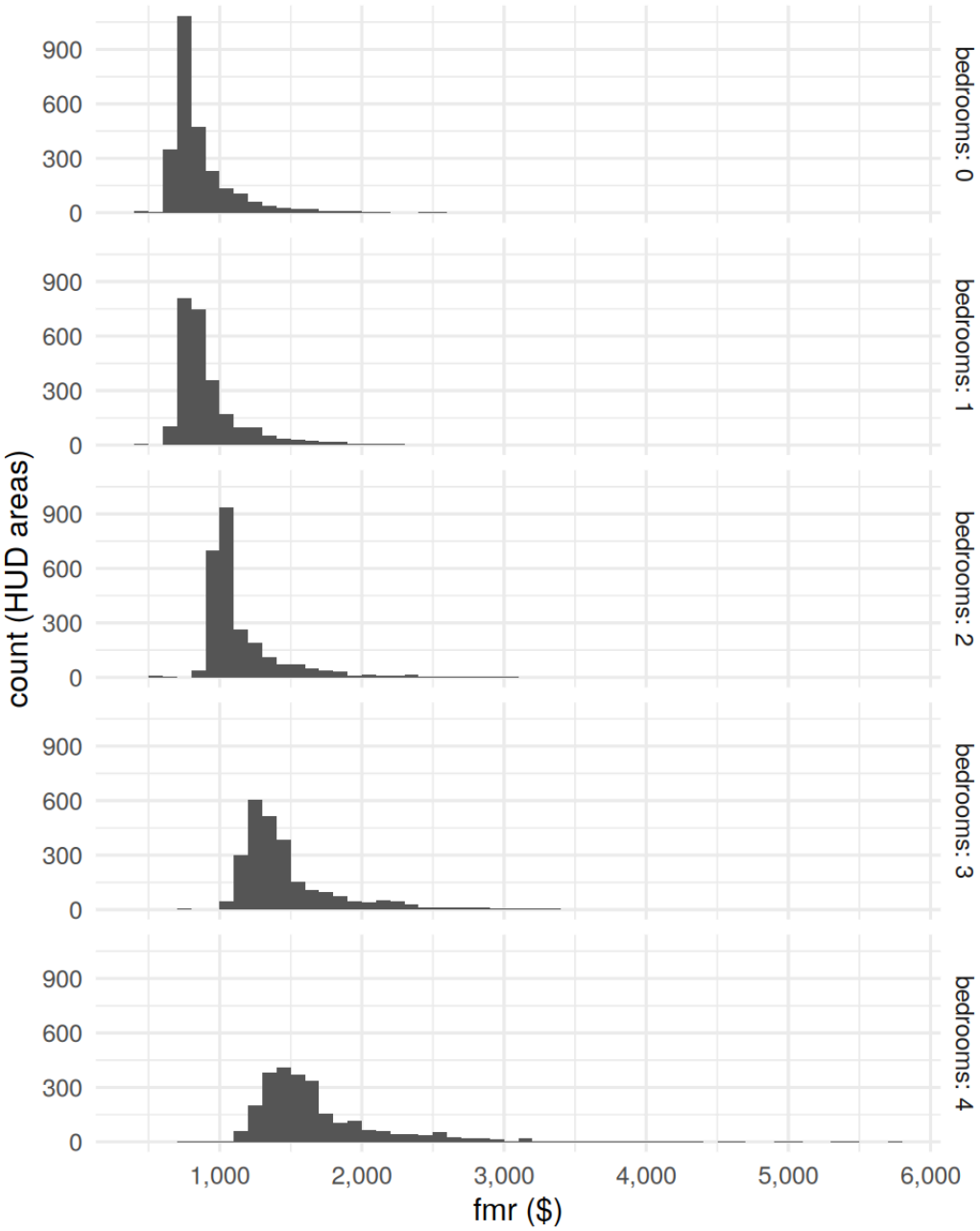


Figure 2

```
ggplot(areas, aes(factor(bedrooms), fmr, fill = metro)) +
  geom_boxplot(outlier.size = 0.8, outlier.alpha = 0.4) +
  scale_y_continuous(labels = scales::label_comma()) +
  labs(x = "bedrooms", y = "fmr ($)", caption = "Figure 3") +
  theme_minimal(12)
```

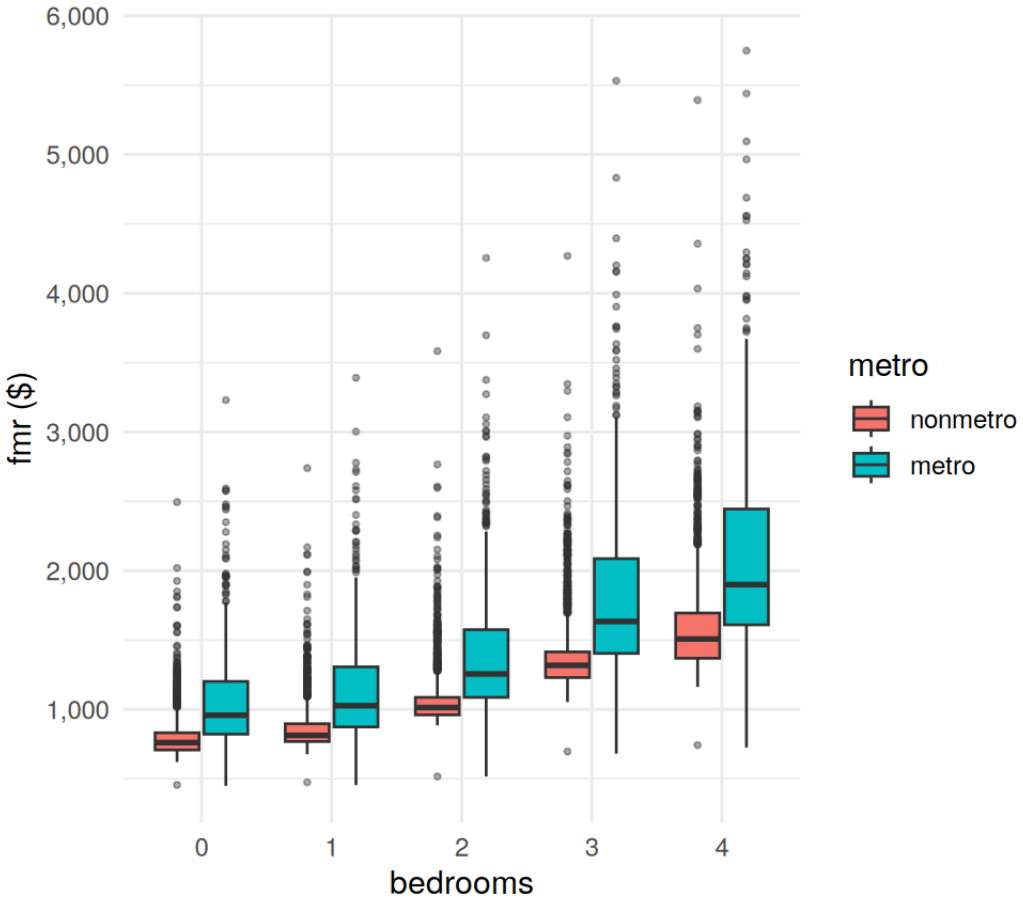


Figure 3

```
# ratio to the 2 bedroom fmr of the same area
ratios <- areas |>
  group_by(hud_area_code) |>
  mutate(ratio = fmr / fmr[bedrooms == 2]) |>
  ungroup() |>
  filter(bedrooms != 2)

ggplot(ratios, aes(ratio)) +
  geom_histogram(binwidth = 0.01, boundary = 0) +
  facet_wrap(~bedrooms, scales = "free_x", labeller = label_both) +
  labs(x = "fmr / fmr (2 bedrooms)", y = "count (HUD areas)",
       caption = "Figure 4") +
  theme_minimal(12)
```

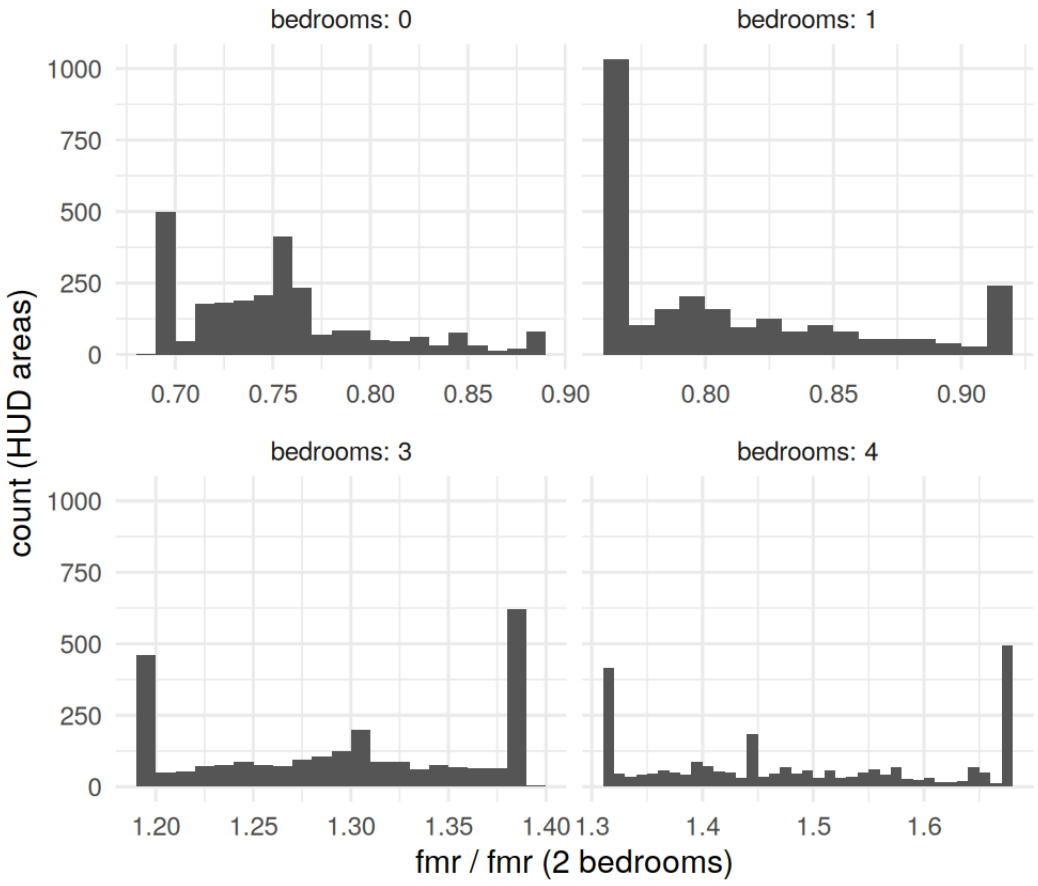


Figure 4

```
state_2 <- state_areas |>
  filter(bedrooms == 2) |>
  mutate(stusps = fct_reorder(stusps, fmr, .fun = median))

ggplot(state_2, aes(fmr, stusps)) +
  geom_point(
    aes(color = metro), alpha = 0.6, size = 1.2,
    position = position_jitter(width = 0, height = 0.25, seed = 1)
  ) +
  stat_summary(fun = median, geom = "point", shape = 124, size = 4) +
  scale_x_continuous(labels = scales::label_comma()) +
  labs(x = "fmr ($)", y = "stusps", caption = "Figure 5") +
  theme_minimal(12)
```

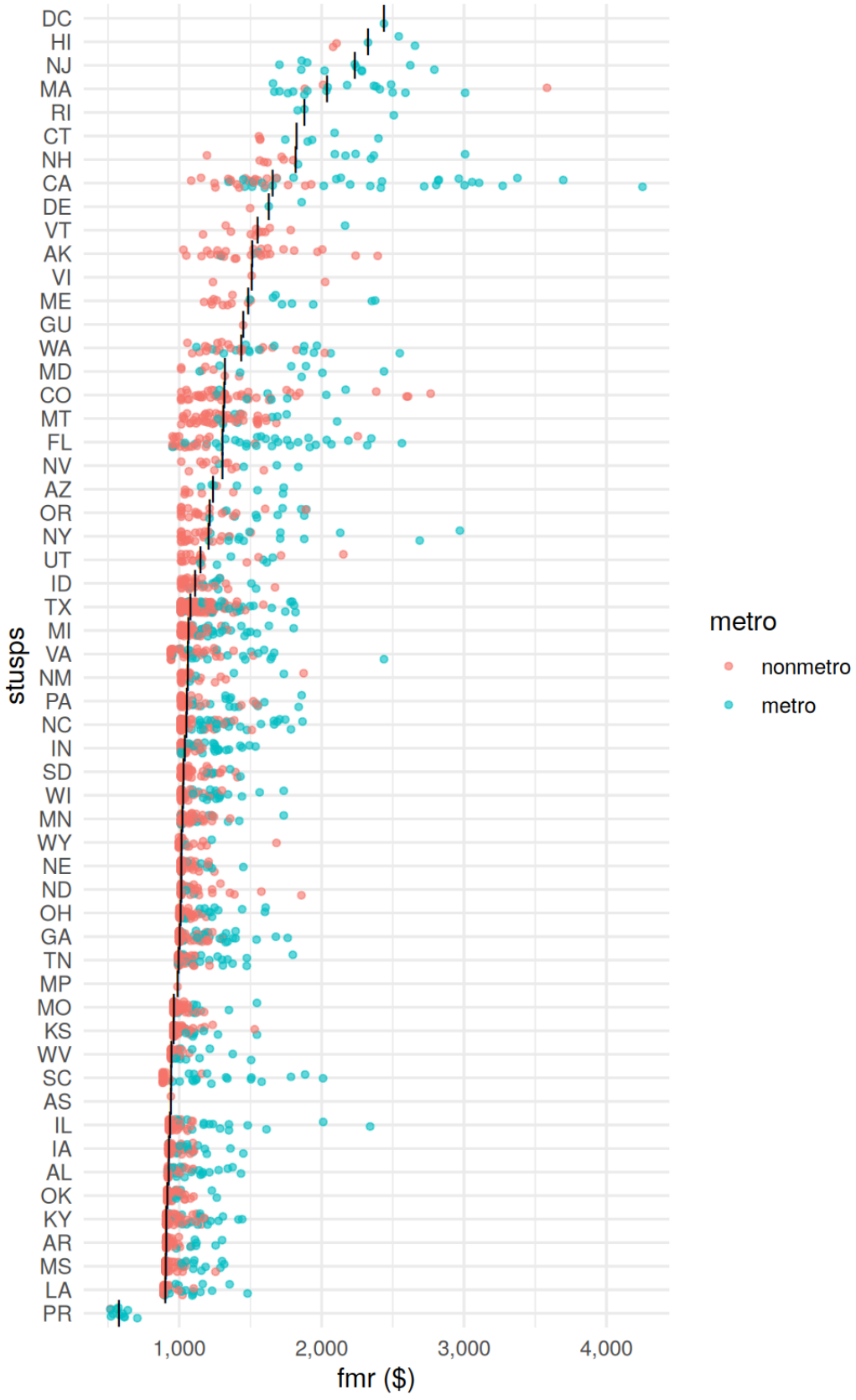


Figure 5

```
ggplot(areas_2, aes(pop, fmr, color = metro)) +
  geom_point(alpha = 0.4, size = 1) +
  scale_x_log10(labels = scales::label_comma()) +
  scale_y_continuous(labels = scales::label_comma()) +
```

```
  labs(x = "pop2024", y = "fmr ($)",
       caption = "Figure 6") +
  theme_minimal(12)
```

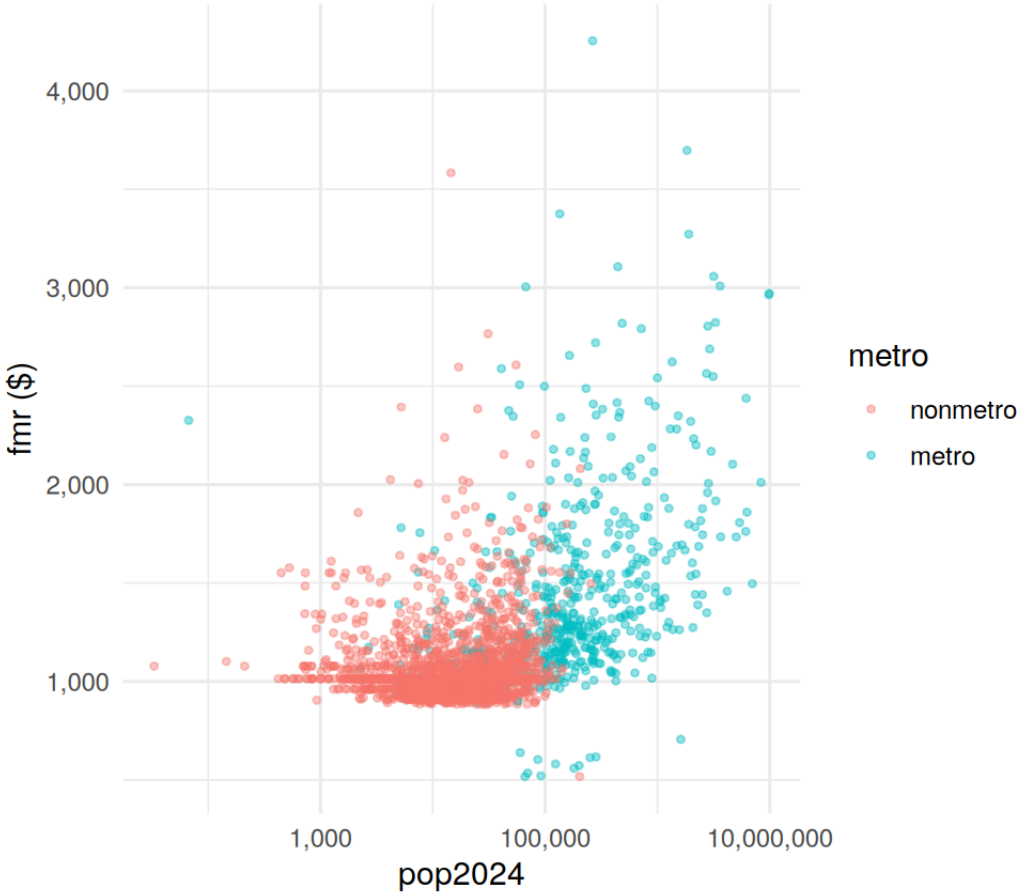


Figure 6

```
# add 1 so the pop2024 = 0 records survive the log
ggplot(fmr_2, aes(pop2024 + 1)) +
  geom_histogram(bins = 40) +
  facet_grid(rows = vars(record_type), labeller = label_both) +
  scale_x_log10(labels = scales::label_comma()) +
  labs(x = "pop2024 + 1", y = "count (records)", caption = "Figure 7") +
  theme_minimal(12)
```

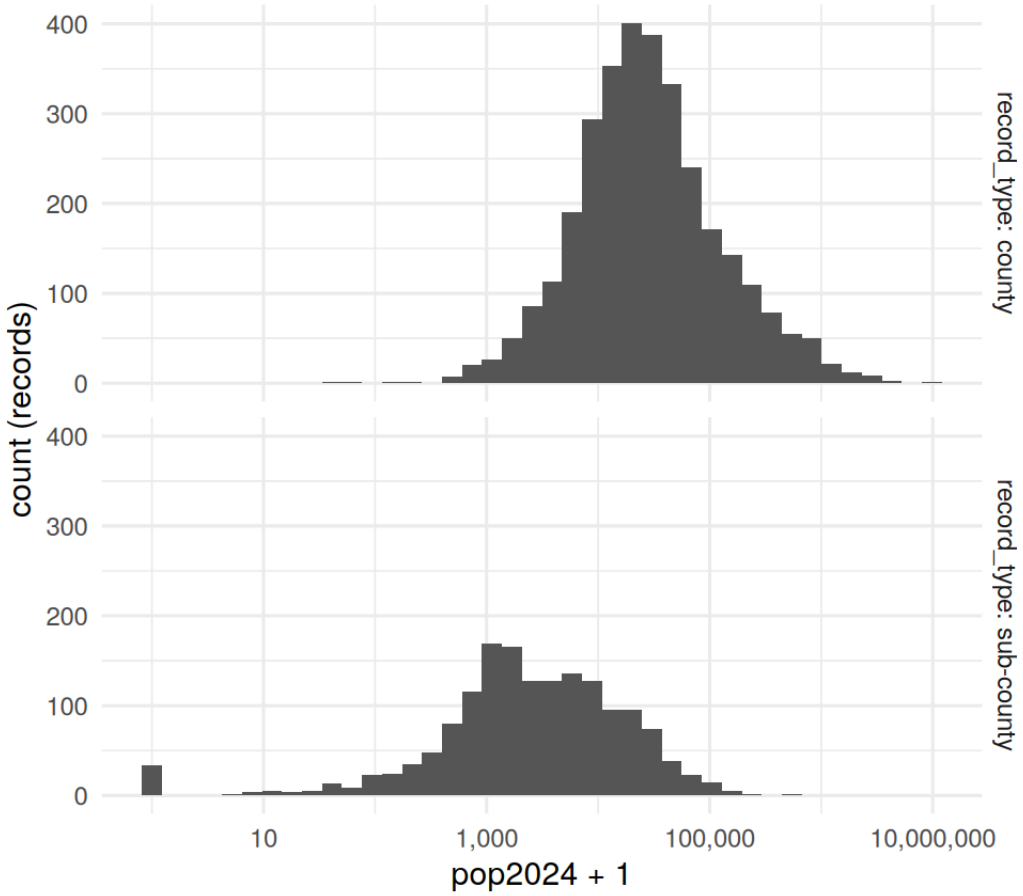


Figure 7

```r
# 2 bedroom fmr by record and by HUD area
units <- bind_rows(
  transmute(fmr_2, unit = "records", fmr),
  transmute(areas_2, unit = "HUD areas", fmr)
)

ggplot(units, aes(fmr, color = unit)) +
  geom_density() +
  scale_x_continuous(labels = scales::label_comma()) +
  labs(x = "fmr ($)", y = "density", caption = "Figure 8") +
  theme_minimal(12)
```

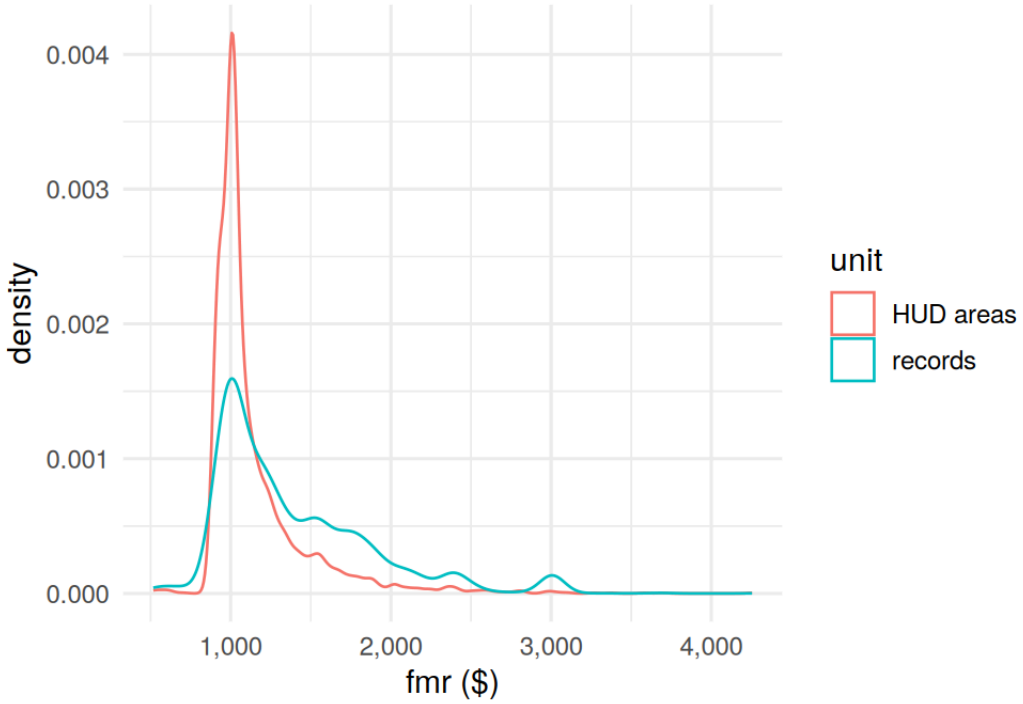


Figure 8

```r
# 12 cheapest and 12 most expensive areas (2 bedrooms)
ranked <- arrange(areas_2, fmr)
extreme_codes <- c(head(ranked$hud_area_code, 12),
                   tail(ranked$hud_area_code, 12))

areas |>
  filter(hud_area_code %in% extreme_codes) |>
  mutate(
    hud_area_name = str_remove(hud_area_name, " HUD Metro FMR Area"),
    hud_area_name = fct_reorder(hud_area_name, fmr)
  ) |>
  ggplot(aes(fmr, hud_area_name, color = factor(bedrooms))) +
  geom_point(size = 2) +
  scale_color_viridis_d() +
  scale_x_continuous(breaks = scales::breaks_width(2000),
                     labels = scales::label_comma()) +
  labs(x = "fmr ($)", y = "hud_area_name", color = "bedrooms",
       caption = "Figure 9") +
  theme_minimal(12)
```

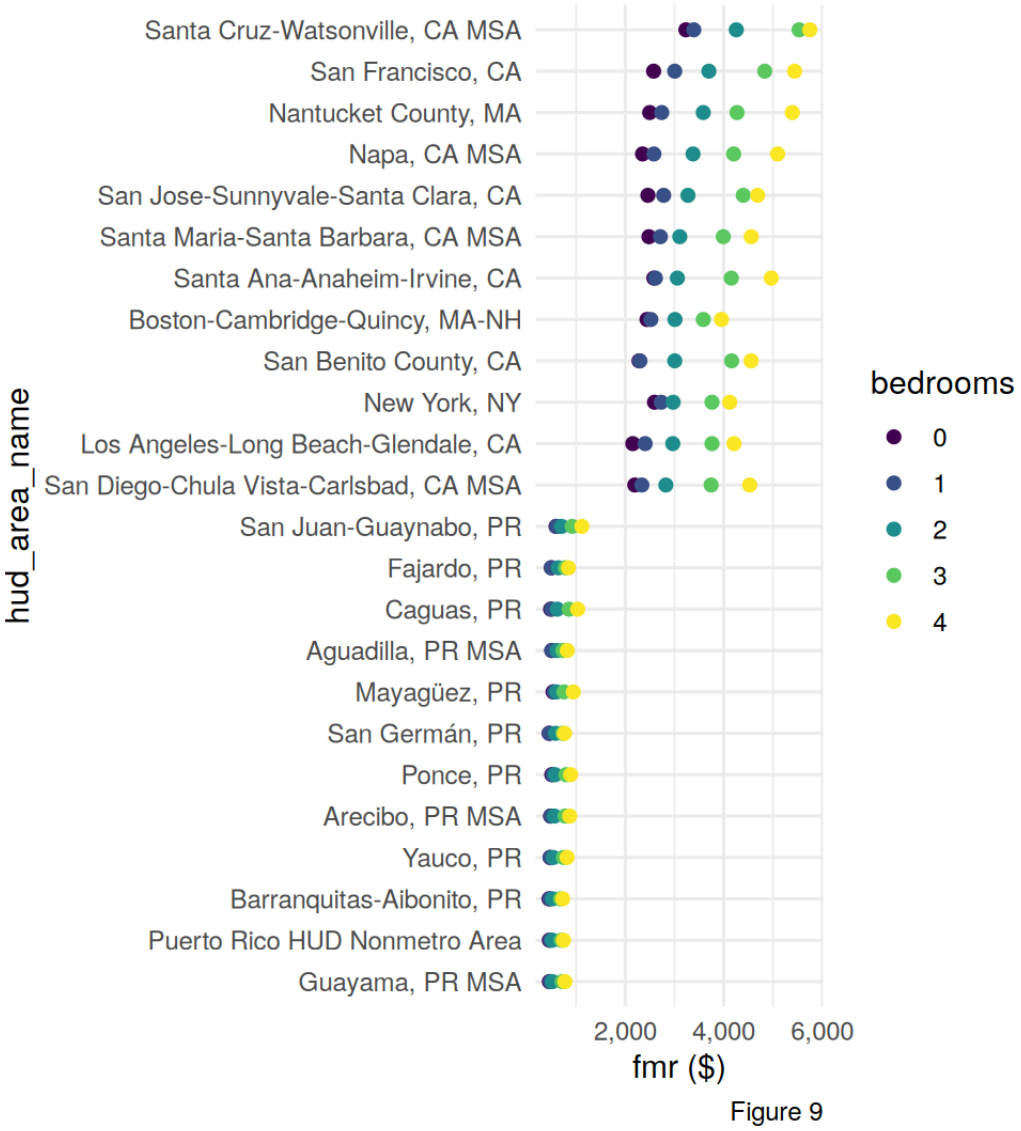


Figure 9

```r
# areas with the most records
areas_2 |>
  slice_max(n_records, n = 15) |>
  mutate(hud_area_name = fct_reorder(hud_area_name, n_records)) |>
  ggplot(aes(n_records, hud_area_name)) +
  geom_point() +
  expand_limits(x = 0) +
  labs(x = "n_records", y = "hud_area_name", caption = "Figure 10") +
  theme_minimal(12)
```

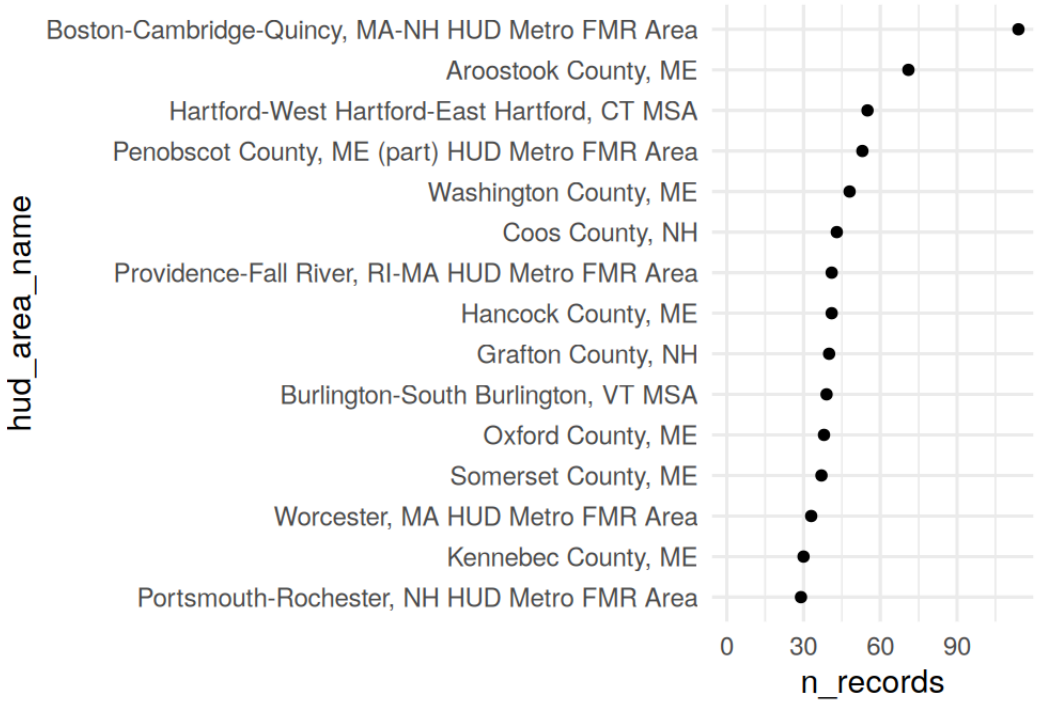

Boston-Cambridge-Quincy, MA-NH HUD Metro FMR Area
Aroostook County, ME
Hartford-West Hartford-East Hartford, CT MSA
Penobscot County, ME (part) HUD Metro FMR Area
Washington County, ME
Coos County, NH
Providence-Fall River, RI-MA HUD Metro FMR Area
Hancock County, ME
Grafton County, NH
Burlington-South Burlington, VT MSA
Oxford County, ME
Somerset County, ME
Worcester, MA HUD Metro FMR Area
Kennebec County, ME
Portsmouth-Rochester, NH HUD Metro FMR Area
hud_area_name
0
30
60
90
n_records

Figure 10

Data checks behind the figures

Checks on the data behind the figures. A record is one `fips`, an area is one `hud_area_code`.

```
cat("rows:", nrow(fmr), " columns:", ncol(fmr) - 1,
    "\nrecords (fips):", n_distinct(fmr$fips),
    "\nHUD areas:", n_distinct(fmr$hud_area_code),
    "\nstates / territories:", n_distinct(fmr$stusps),
    "\nbedroom sizes:", paste(sort(unique(fmr$bedrooms)), collapse = ", "),
    "\n")
```

```
rows: 23820  columns: 11
records (fips): 4764
HUD areas: 2606
states / territories: 56
bedroom sizes: 0, 1, 2, 3, 4
```

```
fmr |>
  select(-record_type) |>
  summarise(across(everything(), ~ sum(is.na(.x)))) |>
  pivot_longer(everything(), names_to = "column", values_to = "n_missing") |>
  filter(n_missing > 0)
```

```
# A tibble: 1 x 2
  column           n_missing
  <chr>                <int>
1 county_town_name     15795
```

```
fmr_2 |>
  count(record_type, missing_town_name = is.na(county_town_name))
```

```
# A tibble: 3 x 3
  record_type missing_town_name     n
  <chr>       <lgl>             <int>
1 county      FALSE                 2
2 county      TRUE               3159
3 sub-county  FALSE              1603
```

```
cat("fully duplicated rows:", sum(duplicated(fmr)),
    "\nduplicated fips + bedrooms:",
    sum(duplicated(fmr[c("fips", "bedrooms")])),
    "\nrecords without every bedroom size:",
    sum(count(fmr, fips)$n != n_distinct(fmr$bedrooms)),
    "\nrecords where fmr ever goes down as bedrooms go up:",
    fmr |>
      arrange(fips, bedrooms) |>
      group_by(fips) |>
      summarise(down = any(diff(fmr) < 0)) |>
      pull(down) |>
      sum(),
    "\n")
```

```
fully duplicated rows: 0
duplicated fips + bedrooms: 0
records without every bedroom size: 0
records where fmr ever goes down as bedrooms go up: 0
```

```
# same fmr within an area?
fmr |>
  group_by(hud_area_code, bedrooms) |>
  summarise(n_fmr = n_distinct(fmr), .groups = "drop") |>
  count(n_fmr)
```

```
# A tibble: 1 x 2
  n_fmr     n
  <int> <int>
1     1 13030
```

```
# records per area, areas in more than one state
areas_2 |>
  summarise(
    areas = n(),
    one_record = sum(n_records == 1),
    two_or_more = sum(n_records > 1),
    most_records = max(n_records)
  )
```

```
# A tibble: 1 x 4
  areas one_record two_or_more most_records
  <int>      <int>       <int>        <int>
1  2606       2325         281          114
```

```
state_areas |>
  filter(bedrooms == 2) |>
  count(hud_area_code) |>
  summarise(areas_in_more_than_one_state = sum(n > 1))
```

```
# A tibble: 1 x 1
  areas_in_more_than_one_state
                         <int>
1                           36
```

```
# METRO codes should be metro = 1, NCNTY codes metro = 0
fmr_2 |>
  mutate(code_start = str_extract(hud_area_code, "^[A-Z]+")) |>
  count(metro, code_start)
```

```
# A tibble: 2 x 3
  metro    code_start     n
  <fct>    <chr>      <int>
1 nonmetro NCNTY       2788
2 metro    METRO       1976
```

```
fmr_2 |>
  filter(pop2024 == 0) |>
  count(stusps, metro)
```

```
# A tibble: 4 x 3
  stusps metro        n
  <chr>  <fct>    <int>
1 ME     nonmetro     8
```

```
2 ME     metro        3
3 NH     nonmetro    17
4 VT     nonmetro     6
```

```
fmr_2 |>
  filter(pop2024 == 0) |>
  select(stusps, countyname, county_town_name) |>
  head(6)
```

```
# A tibble: 6 x 3
  stusps countyname       county_town_name
  <chr>  <chr>            <chr>
1 ME     Aroostook County Penobscot Indian Island Reservation
2 ME     Franklin County  West Central Franklin UT
3 ME     Hancock County   Marshall Island UT
4 ME     Hancock County   Northwest Hancock UT
5 ME     Knox County      Criehaven UT
6 ME     Knox County      Muscle Ridge Island UT
```

```
fmr_2 |>
  count(stusps, record_type) |>
  filter(record_type == "sub-county")
```

```
# A tibble: 6 x 3
  stusps record_type     n
  <chr>  <chr>       <int>
1 CT     sub-county    169
2 MA     sub-county    351
3 ME     sub-county    529
4 NH     sub-county    259
5 RI     sub-county     39
6 VT     sub-county    256
```

```
fmr |>
  group_by(bedrooms) |>
  summarise(min = min(fmr), median = median(fmr), mean = round(mean(fmr)),
            max = max(fmr))
```

```
# A tibble: 5 x 5
  bedrooms   min median  mean   max
     <dbl> <dbl>  <dbl> <dbl> <dbl>
1        0   450   933   1050  3230
2        1   455   978.  1125  3390
3        2   517  1228   1388  4254
4        3   681  1611   1784  5531
5        4   726  1850.  2028  5748
```

```
# most common 2 bedroom fmr
areas_2 |>
  count(fmr, sort = TRUE) |>
  head(5)
```

```
# A tibble: 5 x 2
    fmr     n
  <dbl> <int>
```

```
1  1014   372
2   909    98
3  1002    67
4   961    66
5   962    62
```

```
areas_2 |>
  left_join(distinct(fmr, hud_area_code, stusps), by = "hud_area_code") |>
  filter(stusps %in% c("AS", "GU", "MP", "VI", "DC")) |>
  select(stusps, hud_area_name, fmr, pop)
```

```
# A tibble: 7 x 4
  stusps hud_area_name                                                fmr    pop
  <chr>  <chr>                                                      <dbl>  <dbl>
1 DC     Washington-Arlington-Alexandria, DC-VA-MD HUD Metro FMR A~  2438 6.16e6
2 AS     American Samoa                                               942 5.55e4
3 GU     Guam                                                        1449 1.59e5
4 MP     Northern Mariana Islands                                     988 5.39e4
5 VI     St. Croix Island, VI                                        1237 5.32e4
6 VI     St. John Island, VI                                         2025 4.20e3
7 VI     St. Thomas Island, VI                                       1509 5.12e4
```